Undergraduate Chemistry Research Internship

Molecular Dynamics

From Learning the Fundamentals to a Complete Water Simulation

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Youssef Ahmed

Youssef@westlake.edu.cn

Westlake University

Under the supervision

of

Prof. Yiqin Gao and Dr. Manas Mondal

Abstract

This project presents a molecular dynamics (MD) study of liquid water using the **SPONGE/Xponge molecular simulation package**, with the objective of developing a complete MD workflow and applying it to investigate how temperature affects the microscopic structure and dynamics of water. The first stage of the work focused on building, preparing, simulating, and validating a periodic system of 862 TIP3P water molecules at 300 K. This established a reliable simulation and analysis workflow for thermodynamic behavior, molecular geometry, and intermolecular structure. The second stage extended this system into a controlled comparison between liquid water at **300 K and 273 K**, using matched simulation and analysis conditions so that the influence of temperature could be examined directly. The comparative analysis shows clear temperature-dependent changes in the microscopic organization of the liquid. At 273 K, the water exhibited a **more populated geometric hydrogen-bond network**, with the average number of hydrogen bonds increasing from **2.572 to 2.775 per water molecule**. The accepted hydrogen bonds were also slightly shorter and more linear. Radial distribution functions showed enhanced local structural ordering in the cooler system, while the mean tetrahedral order parameter increased from **0.5617 at 300 K to 0.5908 at 273 K**. In contrast, mean density remained very similar between the two systems. Mean-squared-displacement analysis further showed that molecular mobility decreased upon cooling, with consistently lower oxygen MSD at 273 K. Overall, the results demonstrate that lowering the temperature from 300 K to 273 K produces a **more structured and more tetrahedrally ordered liquid-water network with reduced molecular mobility**, while the system remains liquid. The study also demonstrates a reproducible SPONGE-based workflow for molecular simulation, validation, and structural analysis of water.

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1. Introduction and Objectives

Molecular dynamics simulates the motion of atoms by integrating Newton's equations of motion under an empirical potential-energy function (a force field). For water — a liquid whose properties are governed by a dynamic, constantly rearranging hydrogen-bond network — MD is the natural computational microscope: it resolves, frame by frame, exactly the structural correlations that experiments average over. Water was chosen as the subject of this internship project because it is simultaneously the most familiar and one of the most structurally subtle liquids, and because the TIP3P model is a standard, well-documented benchmark for learning simulation methodology.

The project followed a deliberate progression. First came a learning phase: the statistical-mechanical foundations of MD — integration timesteps, force fields, periodic boundaries, electrostatics, thermostats and barostats — were studied and then exercised hands-on with the SPONGE simulation engine and the Xponge system builder. Second, that knowledge was applied to build, stabilize, and validate a room-temperature (300 K) liquid-water simulation (Experiment 1). Third, a matched temperature comparison (Experiment 2) asked how the structure of the same liquid changes when the target temperature is lowered to 273 K, followed by a deep structural analysis of the hydrogen-bond network. Finally, the analysis itself was subjected to independent review and repair, and the accepted results were packaged with full provenance.

The Experiment-2 research question was precise: for the same TIP3P model, the same number of waters, and the same analysis methodology, what differs between a 300 K branch and a 273 K branch in thermodynamic averages, radial distribution functions, hydrogen-bond population and geometry, tetrahedral order, coordination number, and short-time molecular mobility? The matched design — equal production lengths and equal frame counts — was chosen so that any observed difference could be attributed to temperature rather than to protocol.

Throughout, the guiding discipline was verification: every stage was checkpointed, interrupted runs were recovered through synchronized restarts, analysis defects discovered in review were repaired and re-validated, and the final numbers in this report are read directly from the frozen, hash-verified project package rather than from memory or intermediate notes.

The report follows the chronology of the work: Section 2 condenses the theoretical foundations, Section 3 the computational setup, Sections 4–5 Experiment 1, Sections 6–7 the construction of the matched comparison, Sections 8–9 the analysis methods and results, and Sections 10–14 validation, discussion, limitations, and outlook. An appendix collects parameters, workflow chains, and package provenance.

2. Molecular Dynamics Foundations

This section summarizes the theory actually used by the simulations, at the depth needed to read the results. It condenses the learning-phase material rather than reproducing it slide by slide.

2.1 Newtonian dynamics and integration

MD propagates a classical many-body system by solving Newton's second law. Each atom i experiences a force derived from the potential energy U of the whole configuration,

$$F_i = -\nabla_i U, \quad a_i = \frac{F_i}{m_i} \qquad \text{(Eq. 1)}$$

where $\vec{F}_i$ and $\vec{a}_i$ are the force and acceleration of atom i, m_i is its mass, and ∇_i denotes the gradient with respect to its coordinates. In practice the equations are integrated in discrete steps; this project used a 1 fs timestep, short enough to resolve the fastest relevant motions once the O–H bonds are constrained (Section 2.5). A velocity-Verlet-style scheme advances positions and velocities consistently,

$$r_i(t + \Delta t) = r_i(t) + v_i(t)\Delta t + \frac{\Delta t^2 F_i(t)}{2m_i} \quad \text{(Eq. 2)}$$

with Δt the timestep. The 1 fs choice is a compromise: smaller steps waste computing time, larger steps destabilize the integration — a trade-off that became very concrete when the first velocity initialization attempts failed (Section 4.3).

2.2 Force fields and the TIP3P water model

The force field defines U: bonded terms (bonds, angles) keep molecular geometry sensible, while non-bonded Lennard-Jones and Coulomb terms govern intermolecular structure. TIP3P is a rigid three-site model: one Lennard-Jones centre and a negative partial charge on the oxygen, positive partial charges on the two hydrogens, fixed O–H bond length (0.9572 Å) and H–O–H angle (104.52°). It is computationally cheap and reproduces the qualitative structure of liquid water — which is exactly the regime of this study. The simulated geometry was verified against these ideal values (Section 5.2).

2.3 Periodic boundaries, the minimum image, and electrostatics

A simulation box of ~30 Å side contains of order a thousand molecules; without special treatment its surface would dominate its physics. Periodic boundary conditions (PBC) replicate the box infinitely, and the minimum-image convention computes each pair interaction using the nearest periodic image,

$$\Delta r_{ij} \leftarrow \Delta r_{ij} - L \cdot \mathrm{round}\left(\frac{\Delta r_{ij}}{L}\right) \quad \text{(Eq. 3)}$$

for a cubic box of side L. The same convention is used inside every structural analysis in this report (RDFs, hydrogen bonds, tetrahedral order, coordination, MSD unwrapping). Short-ranged Lennard-Jones and real-space Coulomb interactions were cut off at 8 Å; long-ranged electrostatics were handled by the Particle Mesh Ewald (PME) method, which interpolates charges onto a grid and solves the reciprocal-space part efficiently — essential for a hydrogen-bonded liquid, where electrostatics dominate.

2.4 Ensembles, thermostats, and barostats

Raw Newtonian integration samples the microcanonical (NVE) ensemble. To compare with laboratory conditions one needs controlled temperature (NVT) and pressure (NPT). A thermostat couples the system to a heat bath; a barostat adjusts the volume. The thermostat families studied during the learning phase include velocity rescaling, Andersen (stochastic collisions), Berendsen (exponential relaxation toward the target — simple and robust but not strictly canonical), Langevin-style stochastic thermostats (friction plus random force), and extended-system methods. The barostat families include Berendsen pressure relaxation and the stochastic cell-rescaling method of Bussi and coworkers, which samples the correct NPT distribution.

The practical choices in this project reflect that landscape. The Middle Langevin thermostat was used for heating and NVT work because its stochastic coupling stabilizes difficult starts. For pressure, a Berendsen barostat was used for the early, rough density relaxation, where robustness matters more than rigorous sampling; the final 80 ps equilibration and the production runs used the Bussi stochastic cell-rescaling barostat, which has the correct ensemble properties. This staged strategy — cheap and forgiving first, rigorous last — is standard practice and proved decisive on laptop hardware, where every failed stage costs hours, not seconds. The ensemble caution matters for interpretation too: because a stochastic thermostat was used, dynamical observables are protocol-dependent, which is one reason Section 12 lists transport conclusions as limited.

2.5 Constraining water: SETTLE

TIP3P water is meant to be rigid. Rather than letting O–H bonds vibrate (which would force a sub-femtosecond timestep), the SETTLE algorithm resets each water's geometry analytically after every step. Runtime SETTLE constraints were active in all simulations here, and the final geometry check (Section 5.2) confirms the rigid geometry held to within 10^{-4} Å over production.

3. Computational System and Methods

3.1 System and simulation parameters

Both experiments simulated the same system: 862 TIP3P water molecules (2586 atoms) in a cubic, periodic box of initial side ~30 Å, with atom order O, H1, H2 repeated per water (verified during system construction). Table 1 collects the parameters shared by the experiments.

Table 1: Simulation parameters common to Experiments 1 and 2.

Parameter	Value
Water model	TIP3P (rigid, 3-site) [1]
System size	862 waters, 2586 atoms
Box	cubic, initial side ~30 Å (orthorhombic; fluctuates under NPT)
Timestep	1 fs
Non-bonded cutoff	8 Å
Electrostatics	Particle Mesh Ewald (PME) [2]
Constraints	runtime SETTLE (rigid water) [7]
Thermostat	Middle Langevin (heating, NVT) [4]
Barostats	Berendsen [3] (early pressure relaxation); Bussi stochastic cell rescaling [5, 6] (final NPT equilibration and production)
Engine / builder	SPONGE v2.0.0 [8] / Xponge [9]
Hardware / environment	Windows laptop, conda environment “sponge-cpu” (Python 3.11)

3.2 Workflow and data discipline

Each simulation stage was executed as a separate, scripted SPONGE run that read the accepted endpoint state (coordinates and velocities) of the previous stage and wrote its own restart state, trajectory, box record, and thermodynamic log. Every stage lived in its own timestamped, self-documenting branch directory with command logs, input/output hashes, and a machine-readable acceptance decision; a stage was only built on after its predecessor passed validation. Trajectories store raw float32 coordinates without per-frame headers, with a companion ASCII box file; analyses recomputed everything from these raw records under the minimum-image convention. This discipline is what made the interruption recoveries (Sections 4 and 7) and the final packaged provenance (Appendix) possible.

4. Experiment 1: Building a Validated 300 K Liquid

Experiment 1 turned a randomly packed box of water into a validated ambient-temperature liquid simulation. The stages below follow the accepted state chain; each was validated before the next began.

4.1 System preparation and energy minimization

A 30 Å cubic box was filled with 862 TIP3P waters at roughly liquid density using Xponge, and the force-field files (charges, Lennard-Jones tables, bonds, exclusions) were generated and audited. Energy minimization then removed the overlaps and clashes inevitable in a random packing. The first minimization attempts stalled with localized hot spots — a small number of atom pairs carrying huge forces — which were diagnosed by force-aware analysis and resolved by localized repair and continued minimization until the accepted endpoint was reached. Runtime limits on laptop hardware made minimization itself an iterative, checkpointed process: rather than one long run, minimization advanced through short validated blocks whose endpoints were checkpointed, so an interruption never cost more than one block. The lesson carried forward to every later stage — short, verifiable units of work joined by explicit, hash-checked state files.

4.2 Velocity initialization and low-temperature conditioning

Assigning room-temperature velocities directly to a freshly minimized box injected far too much energy for the rigid-geometry constraints to absorb, and the first attempts failed immediately. The accepted solution was a cold start: a valid rigid-water 50 K seed velocity was generated, the system was quenched to a 10 K target with a very short timestep, then warmed through 25 K and conditioned at 50 K before it was considered heating-ready. This low-temperature conditioning chain was the single most delicate part of Experiment 1.

4.3 Heating to 300 K and NVT stabilization

From the conditioned 50 K state the system was heated in stages to the 300 K target under the Middle Langevin thermostat, then stabilized for 10 ps at constant volume (NVT). Staged heating with a stochastic thermostat avoided the temperature overshoots and integration failures that a single aggressive heating step had produced in earlier attempts. The NVT window also served as a holding pattern in which the system's kinetic and potential energies could be checked for drift before any barostat was allowed to change the box.

4.4 Pressure relaxation, NVE diagnostic, and 80 ps equilibration

Pressure was first relaxed with a forgiving Berendsen barostat (practical 5 ps NPT stage). A short NVE diagnostic then screened for obvious short-time numerical or structural instability without active thermostat or barostat control. Only then was the system handed to the final production-oriented Bussi stochastic-cell-rescaling NPT equilibration stage, during which the density settled from a slightly under-dense start toward its plateau (Figure 1).

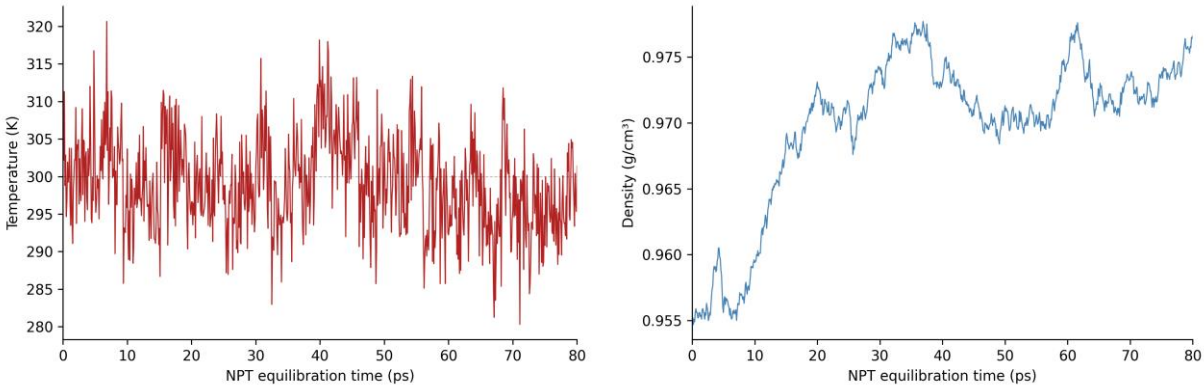


Figure 1: Experiment 1, 80 ps Bussi-NPT equilibration: instantaneous temperature (left) fluctuates around the 300 K target while the density (right) relaxes from ~0.955 g/cm^3 to a plateau near 0.974 g/cm^3.

4.5 Segmented production and recovery from interruption

The production run was planned as a single 10 ps trajectory but was interrupted by a laptop restart after 6 ps. Because every stage wrote restart states and logs, the run was resumed from the last synchronized step as a 4 ps continuation, giving an accepted segmented production of 6 ps + 4 ps. Unsynchronized tail data — frames written after the last checkpoint whose state could not be trusted — were excluded rather than silently concatenated. The accepted primary analysis window is 107.720–112.620 ps (50 thermodynamic records).

5. Experiment 1: Results

5.1 Thermodynamic behaviour

Over the primary production window the system behaved as a stable ambient liquid (Table 2): mean temperature 300.3862 ± 5.6816 K, density 0.974238 ± 0.000979 g/cm^3, potential energy −8203.5226 ± 34.1046 kcal/mol, and box volume 26468.729 ± 32.497 Å^3 (n = 9 recorded frames). Figure 2 shows the production temperature series: stationary fluctuation about the target, with no residual drift from the heating history.

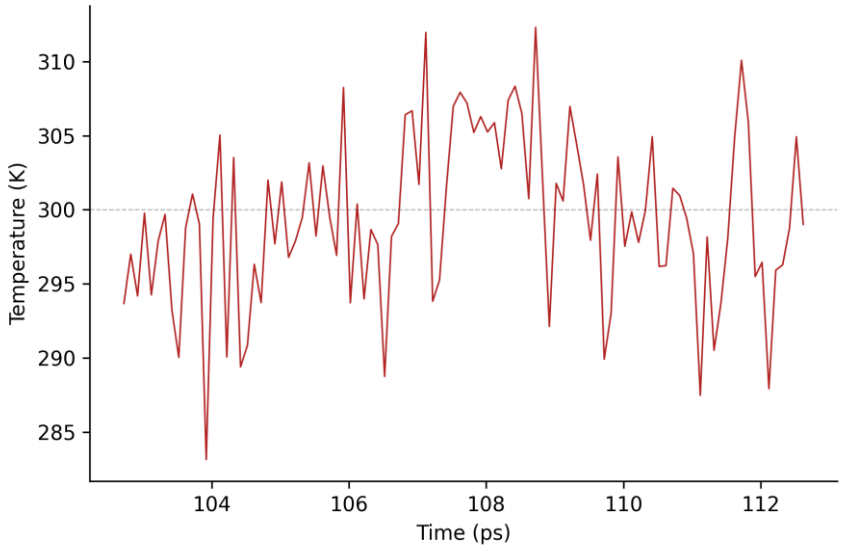


Figure 2: Experiment 1: temperature over the accepted segmented 300 K production (6 ps + 4 ps).

Table 2: Experiment 1 primary thermodynamics (accepted production window).

Quantity (primary window 107.720–112.620 ps)	Mean ± SD
Temperature (n = 50)	300.3862 ± 5.6816 K
Pressure (n = 50)	9.369 ± 483.600 bar
Density (n = 50)	0.974238 ± 0.000979 g/cm^3
Potential energy (n = 50)	−8203.5226 ± 34.1046 kcal/mol
Box volume (n = 9 frames)	26468.729 ± 32.497 Å^3

The pressure row deserves emphasis: instantaneous NPT pressure in a box this small fluctuates by hundreds of bar, and the observed mean of ≈ 9 bar carries an uncertainty of the same order as the target itself. A near-zero mean pressure is therefore not evidence of equilibration; equilibration was instead judged from the stability of temperature, density, and potential energy. This caution carries over directly to Experiment 2, where pressure is reported as descriptive only.

5.2 Rigid geometry check

The runtime SETTLE constraints held the TIP3P geometry essentially exactly: pooled over production, O–H = 0.95719658 ± 0.00004713 Å, H–H = 1.51359458 ± 0.00007450 Å, and H–O–H = 104.490602 ± 0.000091°, against ideal TIP3P values of 0.9572 Å and 104.52°. The integrator-constraint combination is sound.

5.3 Intermolecular structure (illustrative)

Figure 3 shows the intermolecular oxygen–oxygen, oxygen–hydrogen, and hydrogen–hydrogen radial distribution functions computed from 14 production frames. They display the expected liquid signatures: an O–O first peak at 2.75 Å (g ≈ 2.61; first minimum 3.45 Å), an O–H first feature at 1.85 Å (g ≈ 1.24; minimum 2.45 Å), an H–H first feature at 2.45 Å (g ≈ 1.25; minimum 2.95 Å), and long-range means of ≈ 1. With only 14 frames these curves are qualitative illustrations of

liquid structure, not quantitative radar benchmarks; the much better-sampled Experiment-2 analysis supersedes them for any comparison.

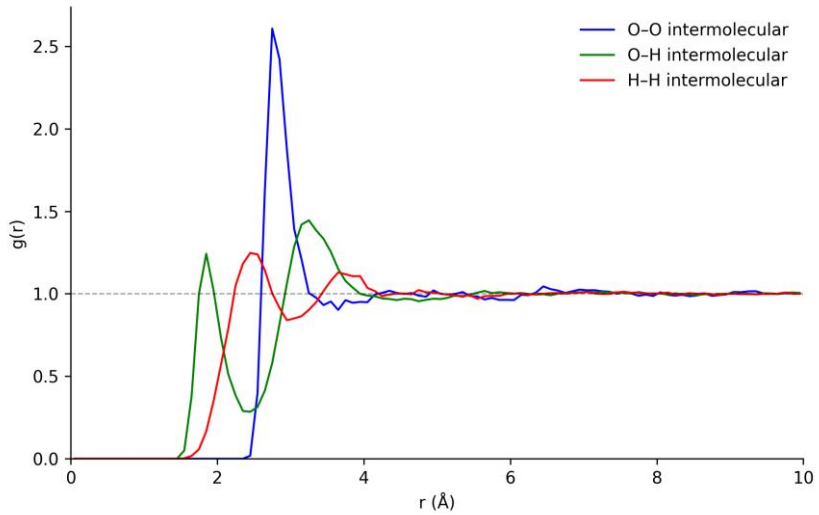


Figure 3: Experiment 1: illustrative 14-frame intermolecular RDFs (O–O, O–H, H–H) of the 300 K liquid. Qualitative only; the matched 100-frame analysis of Experiment 2 is the quantitative authority.

6. Experiment 2: A Matched 300 K / 273 K Design

With a validated 300 K liquid in hand, the project turned to its comparative question: what changes in the structure of the same liquid when the target temperature is lowered to 273 K? To make the comparison honest, everything that could be held fixed was held fixed (Table 3): the same TIP3P model and force-field files, the same 862 waters, the same integration and constraint settings, the same 5 ps production length, the same 0.05 ps frame spacing, and exactly 100 analysis frames per branch processed by the same analysis code. The intended difference is the target temperature alone; the unavoidable residual differences are provenance (the 273 K branch descends from the 300 K lineage by staged cooling) and stochastic history (thermostat random seeds), which no real workflow can eliminate.

Table 3: Matched design of Experiment 2. Only the target temperature and the unavoidable provenance/seed history differ.

Design element	300 K branch	273 K branch
Water model / system	TIP3P, 862 waters, 2586 atoms	identical
Integration	1 fs, PME, 8 Å cutoff, SETTLE	identical
Production ensemble	Bussi NPT, 300 K target	Bussi NPT, 273 K target
Production length	5 ps (dense reference)	5 ps (segmented 3 ps + 2 ps)
Frame spacing / count	0.05 ps / 100 frames	0.05 ps / 100 frames
Analysis frames (time)	112.670–117.620 ps	124.670–129.620 ps
Structural / box / thermo records	100 / 100 / 100	100 / 100 / 99 (one record genuinely lost, not fabricated)
Analysis code & criteria	shared, hash-pinned	identical

The 300 K side is a dense reference segment cut from the validated Experiment-1 lineage. The 273 K side had to be manufactured by controlled cooling — the subject of Section 7.

7. Building the 273 K Dataset

7.1 The 300 K dense reference

The 300 K branch of the comparison is a dedicated dense-reference run: 100 structural frames spaced 0.05 ps over 112.670–117.620 ps, with synchronized box and thermodynamic records (100 each), validated and hash-pinned before use. Because it descends from the validated Experiment-1 lineage, no new equilibration question arose on the 300 K side; the methodological risk concentrated entirely on manufacturing a trustworthy 273 K liquid, which is what the rest of this section describes.

7.2 Staged cooling: 300 → 290 → 280 → 273 K

A liquid cannot simply be re-targeted to a lower temperature without consequence: changing the thermostat set-point to 273 K does not make the instantaneous temperature 273 K — the kinetic energy must drain through the thermostat over a finite relaxation time, and a single brutal quench risks the same kind of instability that the cold-start conditioning had been designed to avoid.

The system was therefore cooled in stages — 300 → 290 → 280 → 273 K — with a validated checkpoint at each step. The checkpoint discipline paid off immediately: an interruption during cooling was recovered by restarting from the last accepted checkpoint state rather than from scratch. A thermal audit of the cooled state (checking that the instantaneous temperature distribution had actually settled at the 273 K target) then motivated one additional 1 ps equilibration stage at 273 K before production was allowed to begin. Only after that stage passed acceptance did the 273 K production run start.

7.3 Segmented production and the synchronized step-3000 restart

The 273 K production was itself interrupted by a second laptop restart. The first segment covered steps 1–3000 (60 frames, 124.670–127.620 ps); the run was resumed from the synchronized step-3000 restart state for a second segment of 40 frames (127.670–129.620 ps). The two segments concatenate cleanly at the frame level. One casualty could not be recovered: the thermodynamic log record at the restart boundary (step 3000, time 127.620 ps) was buffered in memory when the machine stopped and is genuinely absent. It was not fabricated; the 273 K branch therefore has 100 structural frames, 100 box records, and 99 thermodynamic records, and every 273 K thermodynamic statistic in this report is computed over n = 99.

8. Deep Analysis Methods

All Experiment-2 analyses share the same loading conventions: raw float32 frames, one box record per frame, minimum-image distances under the exact per-frame box, and identical code for both temperature branches.

8.1 Thermodynamics

Temperature, pressure, density, and potential energy were parsed from the run logs (header-driven parsing, no assumed column order), and box volumes were computed exactly as Lx·Ly·Lz from the 100 synchronized box records per branch. Means and sample standard deviations over the production frames are reported; pressure is descriptive only, for the reasons given in Section 5.1.

8.2 Radial distribution functions

The RDF g(r) measures how the probability of finding a neighbour at distance r compares with a uniform liquid: g(r) > 1 marks preferred separations, g(r) ≈ 1 marks bulk-like disorder, and the position and sharpness of the first peak (and the depth of the first minimum behind it) quantify local structure. Intermolecular O–O, O–H, and H–H RDFs were accumulated with 0.05 Å bins using minimum-image distances and exact per-frame volumes, normalized against the same ideal-gas reference for both branches so that the curves are directly comparable. Feature detection used a light 3-bin smoother to locate extrema while reporting raw g values; a "later robust feature" means a larger-distance feature that repeatably clears prominence and fluctuation thresholds — i.e. structure, not noise.

8.3 Hydrogen bonds

A geometric hydrogen-bond definition (Figure 4) was used identically in both branches and in the network visualization of Section 9.8. A donor water D and an acceptor water A (different molecules) are counted as hydrogen-bonded when

donor–acceptor O···O distance < 3.5 Å AND angle O_D–H···O_A > 150° (vertex at H),

evaluated under the minimum-image convention and counted as donor-O/H/acceptor-O triplet events (each donor hydrogen tested separately). The mean events per frame and bonds per water follow directly; the accepted distance and angle distributions are reported in Section 9.3.

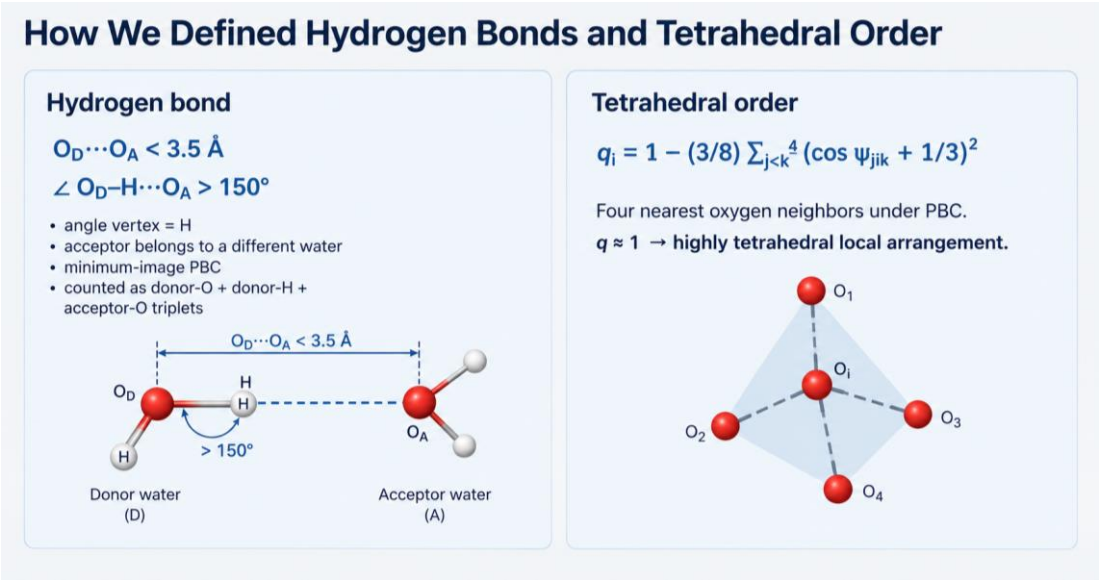


Figure 4: Definitions used for the network analysis: the geometric hydrogen-bond criterion (left) and the tetrahedral order parameter (right). Schematic from the project presentation; criteria identical to the analysis code.

8.4 Tetrahedral order parameter

To measure how tetrahedrally ordered each molecule's local environment is, the standard tetrahedral order parameter [10] was computed from the four nearest oxygen neighbours of each water,

$$q_i = 1 - \frac{3}{8}\sum_{j<k}^{4}\left(\cos\psi_{jik} + \frac{1}{3}\right)^2 \qquad \text{(Eq. 4)}$$

where ψ_jik is the angle formed by neighbours j and k with water i at the vertex. A perfect tetrahedron gives q = 1; a random arrangement gives q ≈ 0. Raw per-water q values were never clipped; the reported means pool 862 waters × 100 frames per branch.

8.5 Coordination at a matched radius

First-shell coordination counts the oxygen neighbours within a cutoff of each oxygen. Because the two branches have slightly different RDF first minima, counting at each branch's own minimum would compare differently sized shells; the comparison therefore uses a matched 3.5 Å cutoff (and cross-evaluated radii) so the two numbers mean the same thing.

8.6 Mean squared displacement and short-time mobility

Translational mobility was quantified by the oxygen mean squared displacement,

$$\mathrm{MSD}(\tau) = \frac{1}{N}\sum_{i}^{N}\langle|r_i(t+\tau) - r_i(t)|^2\rangle \qquad \text{(Eq. 5)}$$

averaged over waters i and over multiple time origins t at lag τ. Trajectories were unwrapped across periodic boundaries before differencing; a centre-of-mass drift correction was applied (its size was measured and found small, ≤ 0.17 Å^2 at 2 ps); and the segmented 273 K production was handled in a restart-safe way — no time origin may straddle the segment boundary, and beyond 1.95 ps only the longer first segment contributes. The Einstein relation connects the long-time slope of the MSD to the self-diffusion coefficient,

$$D = \frac{1}{6}\frac{d\,\mathrm{MSD}}{dt} \quad (\text{long} - \text{time linear regime}) \qquad \text{(Eq. 6)}$$

but that regime is exactly what a 5 ps trajectory cannot demonstrate. Candidate fitting windows were scored against prospective reliability criteria (positive slope, high R^2, diffusive exponent near 1, low curvature, stable local slopes, enough lag points, and enough valid 273 K origins). No common fitting window satisfied all criteria for both temperatures, so the accepted result contains no fitted value: no converged bulk diffusion coefficient is claimed anywhere in this report.

9. Results: 300 K versus 273 K

Every number in this section is read from the accepted analysis outputs inside the final project package; figures are the final validated renderings.

9.1 Thermodynamics

The two branches sample their targets as expected (Table 4, Figure 5): 301.7867 ± 5.3031 K (n = 100) at the 300 K target versus 269.9610 ± 4.7604 K (n = 99) at the 273 K target. Density is essentially identical — 0.976319 ± 0.001366 versus 0.976066 ± 0.000786 g/cm^3 — as are the exact box volumes (26409.265 ± 36.923 versus 26415.944 ± 21.317 Å^3): these short trajectories

resolve no density change. Potential energy is markedly lower in the cooler branch (−8214.3944 ± 36.2132 versus −8518.8839 ± 27.1919 kcal/mol), consistent with its more ordered network. Pressure (−75.25 ± 395.94 versus −592.70 ± 357.78 bar) is extremely noisy in this small system and is reported descriptively only; its mean is not evidence of equilibration (Figure 7). Potential energy and exact box volume are shown in Figure 6.

Table 4: Experiment 2 thermodynamics. Means ± SD over production records; the 273 K thermo statistics use n = 99 because one record was genuinely lost at the restart boundary.

Quantity	300 K branch	273 K branch
Sampled temperature	301.7867 ± 5.3031 K (n = 100)	269.9610 ± 4.7604 K (n = 99)
Density	0.976319 ± 0.001366 g/cm^3	0.976066 ± 0.000786 g/cm^3
Potential energy	−8214.3944 ± 36.2132 kcal/mol	−8518.8839 ± 27.1919 kcal/mol
Box volume (exact, n = 100)	26409.265 ± 36.923 Å^3	26415.944 ± 21.317 Å^3
Pressure (descriptive only)	−75.25 ± 395.94 bar	−592.70 ± 357.78 bar

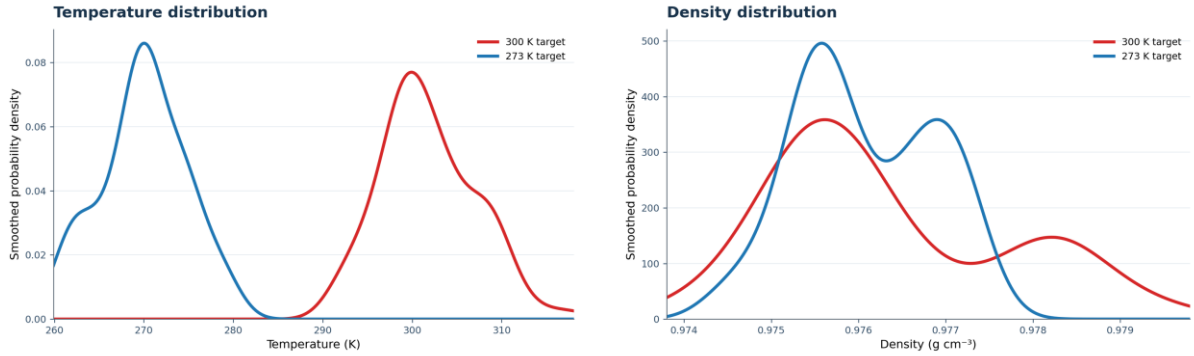


Figure 5: Smoothed sampled-temperature (left) and density (right) distributions for the matched 300 K and 273 K branches. The temperatures clearly occupy different thermal regimes, while the mean densities remain very similar. Kernel-density estimate (KDE) visualization: the KDE is used for presentation clarity; numerical means and SDs are computed from the underlying raw samples.

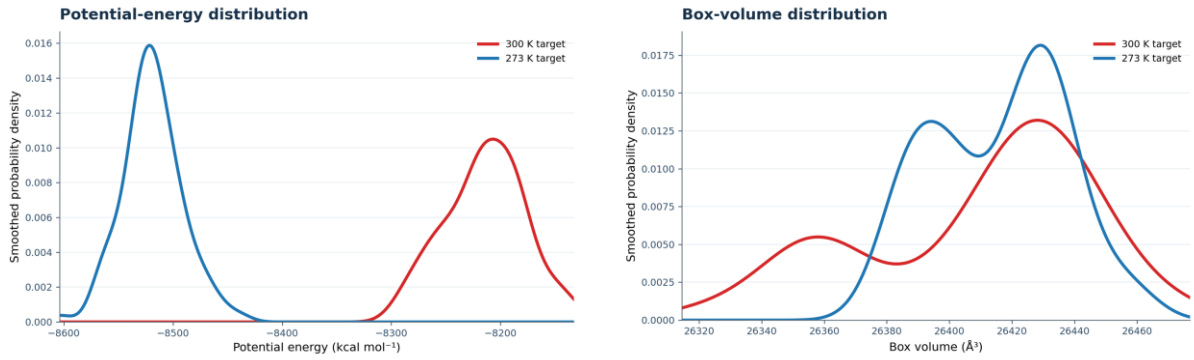


Figure 6: KDE visualization of the sampled potential-energy (left) and exact box-volume (right) distributions. The cooler branch's potential energy is ~305 kcal/mol lower; the volume distributions overlap almost perfectly. KDE is used for presentation clarity; means and SDs are computed from the raw samples.

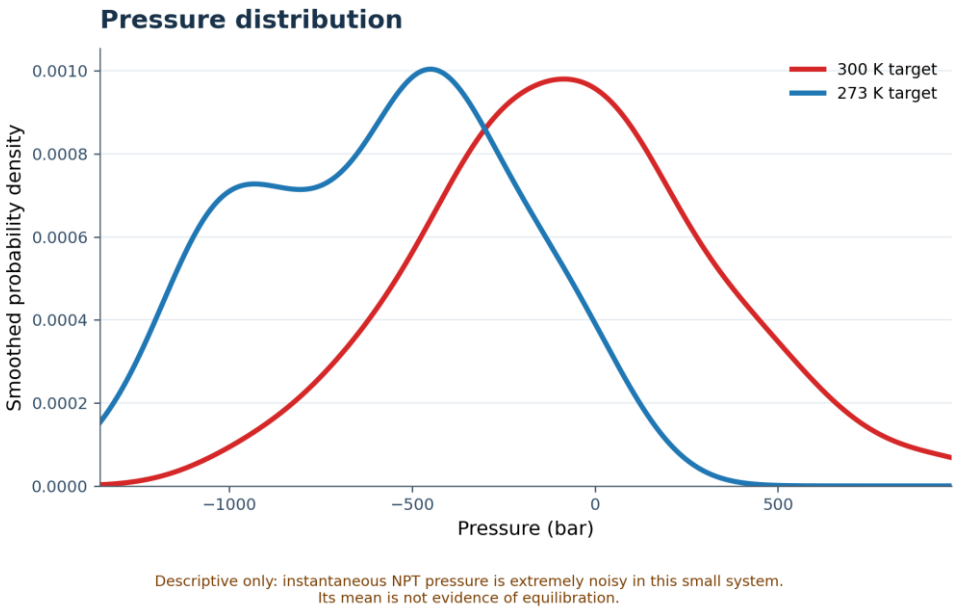


Figure 7: KDE visualization of the instantaneous-pressure distributions of both branches. Instantaneous NPT pressure is extremely noisy in this small system; its mean is descriptive only and is not evidence of equilibration.

9.2 Hydrogen-bond population

The cooler liquid sustains a larger hydrogen-bond network by every population measure: 1108.70 ± 17.00 events per frame at 300 K versus 1195.86 ± 17.19 at 273 K (≈ 7.9% more events), equivalently 2.5724 ± 0.0394 versus 2.7746 ± 0.0399 bonds per water (Figure 8). The two branches fluctuate around well-separated means, and their pooled distributions barely overlap.

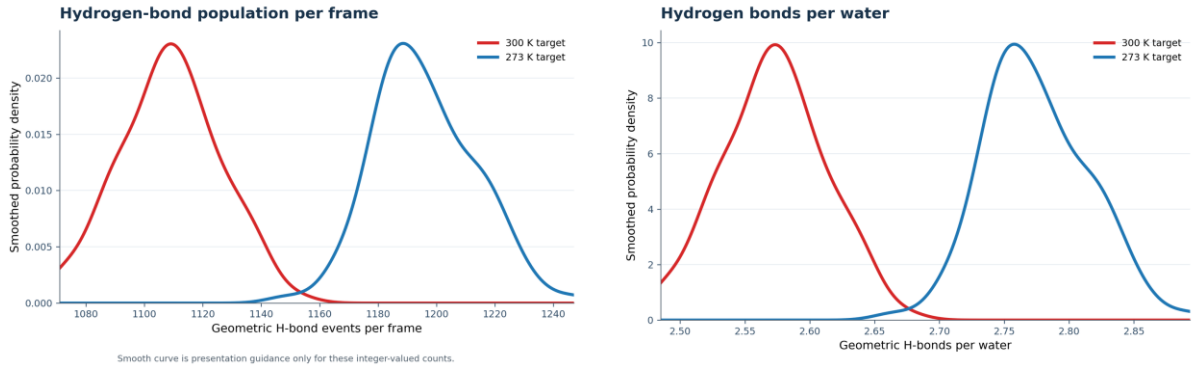


Figure 8: KDE visualization of the per-frame geometric hydrogen-bond count (left) and hydrogen bonds per water (right) for the 100 frames of each branch; the 273 K branch is consistently more populated. The KDE is a smoothed presentation of integer-valued counts; all reported means and SDs are computed from the raw integer samples.

9.3 Hydrogen-bond geometry

Among accepted hydrogen bonds, the donor–acceptor distance is slightly shorter at 273 K (2.8523 ± 0.1809 Å versus 2.8614 ± 0.1910 Å) and the O_D–H···O_A angle slightly straighter (163.393 ± 7.277° versus 162.991 ± 7.311°; Figure 9). The joint distance–angle maps (Figure 10) show the same population shifting toward the short-and-straight corner. These are geometric statements only: geometrically shorter or straighter hydrogen bonds are consistent with, but do not directly measure, stronger hydrogen-bond energies — no hydrogen-bond energy claim is made.

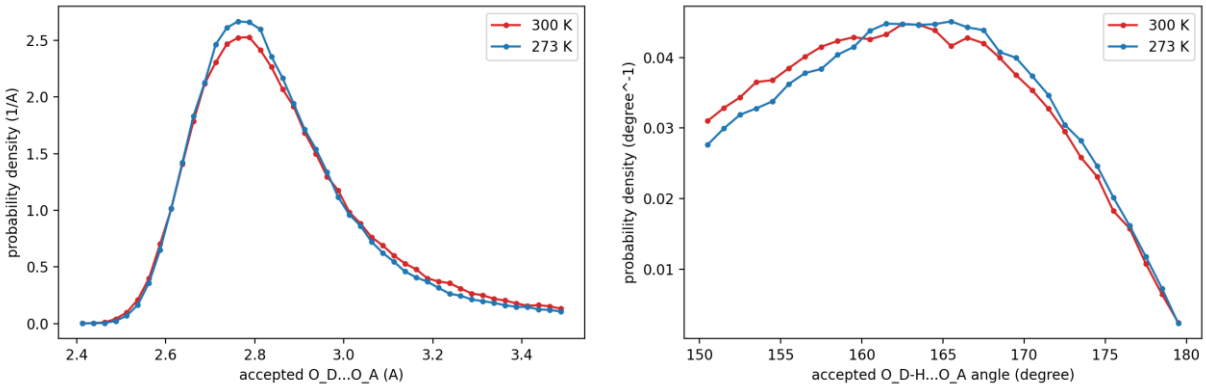


Figure 9: Accepted hydrogen-bond geometry: O···O distance distribution (left) and O_D–H···O_A angle distribution (right) for both branches.

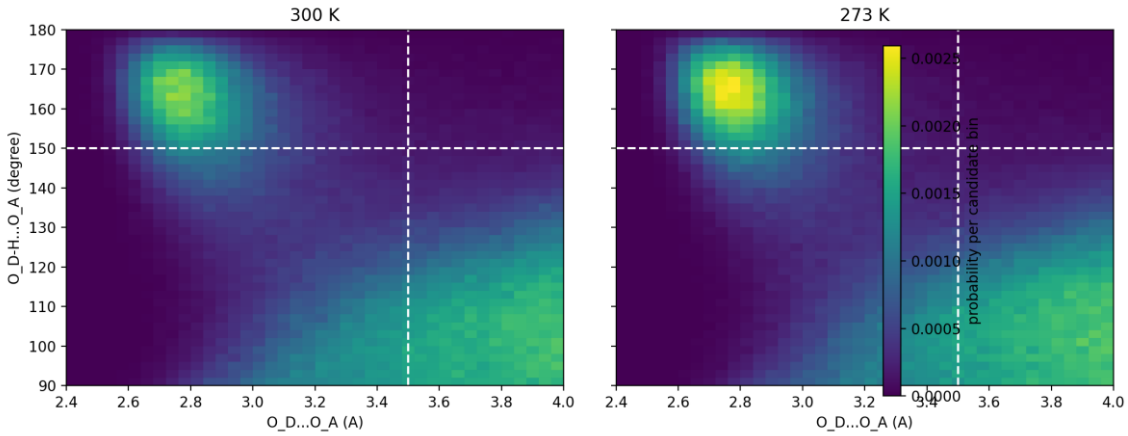


Figure 10: Joint distance–angle probability maps (probability per candidate bin, shared colour scale). Dashed lines mark the acceptance cutoffs (3.5 Å, 150°).

9.4 Radial distribution functions

The matched 100-frame RDFs (Figure 11) show the cooler liquid's structure sharpening without shifting its length scales. The O–O first peak sits at 2.775 Å in both branches but is higher at 273 K (g = 2.921 versus 2.708), with a slightly deeper first minimum. The intermolecular O–H first feature at 1.825 Å is more pronounced at 273 K (g = 1.401 versus 1.256), and both branches show a robust later O–H feature at 3.225 Å (a repeatable structural/orientational contact, distinct from noise). The H–H first feature shifts inward from 2.475 Å to 2.425 Å, and both branches carry a robust later H–H feature near 3.775 Å. No robust later O–O feature was accepted; broad second-shell candidates remained unresolved.

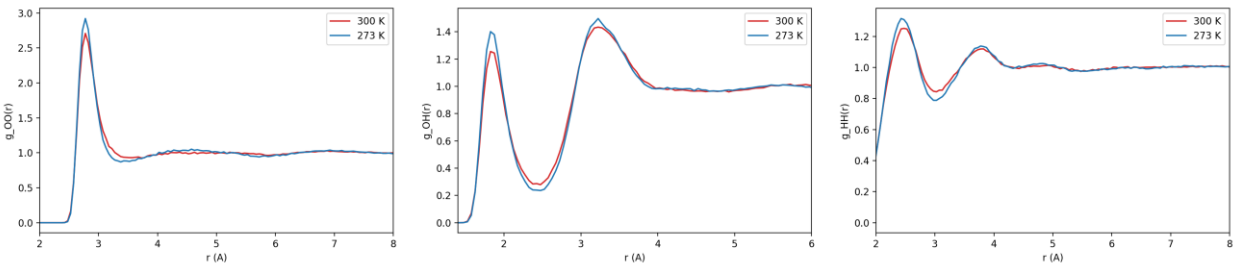


Figure 11: Matched intermolecular RDFs (left to right: O–O, O–H, H–H) for the 300 K and 273 K branches; locations from a light 3-bin smoother, reported g values raw.

9.5 Tetrahedral order

The mean tetrahedral order parameter rises from 0.5617 ± 0.1947 at 300 K to 0.5908 ± 0.1922 at 273 K (pooled raw per-water q; Figure 12). The observed increase is modest: the 273 K liquid is locally more tetrahedrally ordered, not crystalline, and q measures local organization only — never proof of ice.

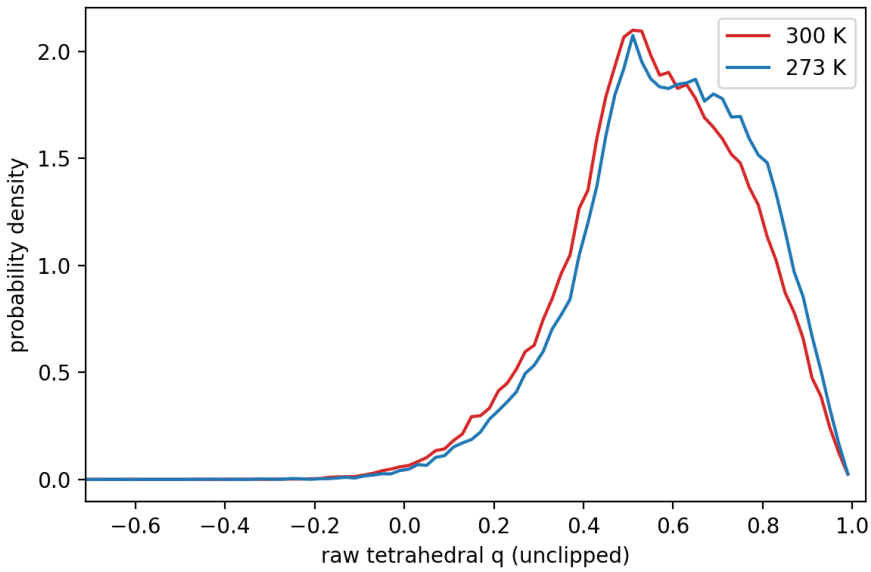


Figure 12: Distribution of the raw per-water tetrahedral order parameter q (862 waters × 100 frames per branch; never clipped).

9.6 Coordination at a matched 3.5 Å radius

Counting oxygen neighbours within the same 3.5 Å cutoff for both branches gives 5.2066 neighbours per oxygen at 300 K versus 5.1170 at 273 K (Figure 13). The change is a small decrease in the number of neighbours within the common 3.5 Å cutoff, indicating only subtle first-shell reorganization — which is why the matched-radius comparison, rather than each branch's own RDF minimum, is the honest like-for-like measure.

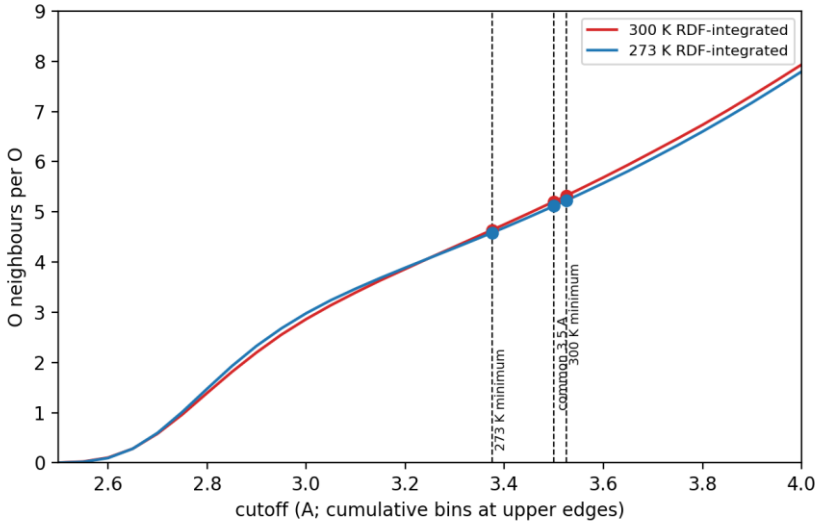


Figure 13: Cumulative O–O coordination as a function of cutoff radius for both branches; the dashed line marks the matched 3.5 Å comparison radius.

9.7 Short-time mobility (MSD)

The oxygen MSD separates cleanly between the branches from the earliest lags (Figure 14): at 1 ps, 3.414 Å^2 (300 K) versus 2.380 Å^2 (273 K); at 2 ps, 6.436 Å^2 versus 4.345 Å^2; at the 2.95 ps limit of the common lag range, 9.404 Å^2 versus 6.266 Å^2. The warmer branch shows consistently higher short-time oxygen MSD — i.e. higher short-time mobility — in these trajectories. For context, selected short-time local fits gave diagnostic apparent short-time estimates of $D \approx 5.17\times10^{-5}$ cm^2/s (300 K) and $D \approx 3.29\times10^{-5}$ cm^2/s (273 K); no common fitting window passed all final reliability criteria, so these are not accepted converged bulk self-diffusion coefficients. Sensitivity checks bound the systematic uncertainties: the centre-of-mass drift correction changes the MSD by at most ~0.17 Å^2 at 2 ps; box-scaling and coordinate-convention choices are negligible (relative slope differences < 10^{-4}); and a restart-consistency test on the 273 K branch changes the MSD at the longest lag by ~2.4%. As detailed in Section 8.6, no common fitting window passed the prospective reliability criteria, so these curves support a mobility comparison only — no converged bulk diffusion coefficient is claimed.

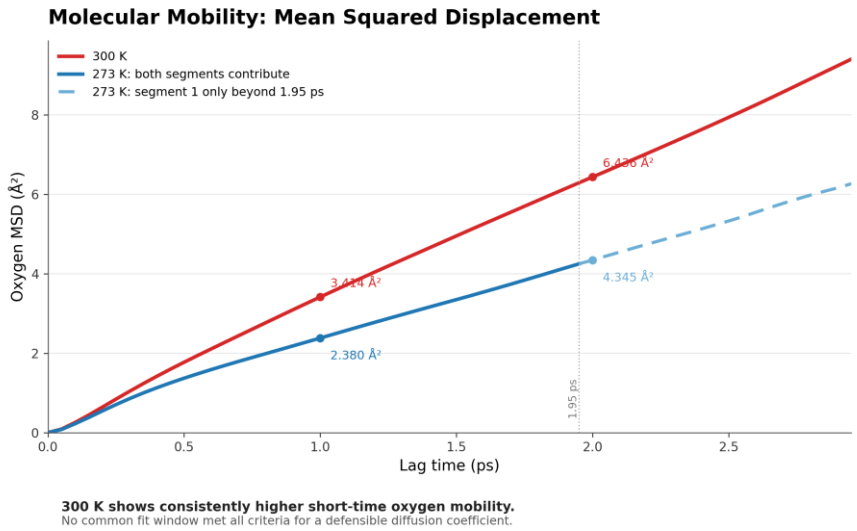


Figure 14: Accepted oxygen MSD comparison (COM-drift corrected, multiple time origins, restart-safe 273 K treatment). The dashed 273 K continuation beyond 1.95 ps is supported by segment 1 only.

9.8 Hydrogen-bond network visualization

To make the network concrete, one objectively chosen frame per branch — the frame whose hydrogen-bond count lies closest to that branch's 100-frame mean (300 K: frame 7, t = 113.020 ps, 1109 whole-frame hydrogen bonds; 273 K: frame 52, t = 127.270 ps, 1196) — was rendered with PyMOL. Each view shows the local cluster around the highest-degree water of that frame (14 waters; dashed lines connect donor-H to acceptor-O of each accepted bond; Figure 15, with close-ups in Figure 16). These local clusters are illustrative only: the drawn clusters carry 15 (300 K) versus 14 (273 K) depicted bonds, so no local connectivity ordering should be read from them — the quantitative conclusion (Section 9.2) rests on the full 100-frame analysis. No ice-formation claim is made.

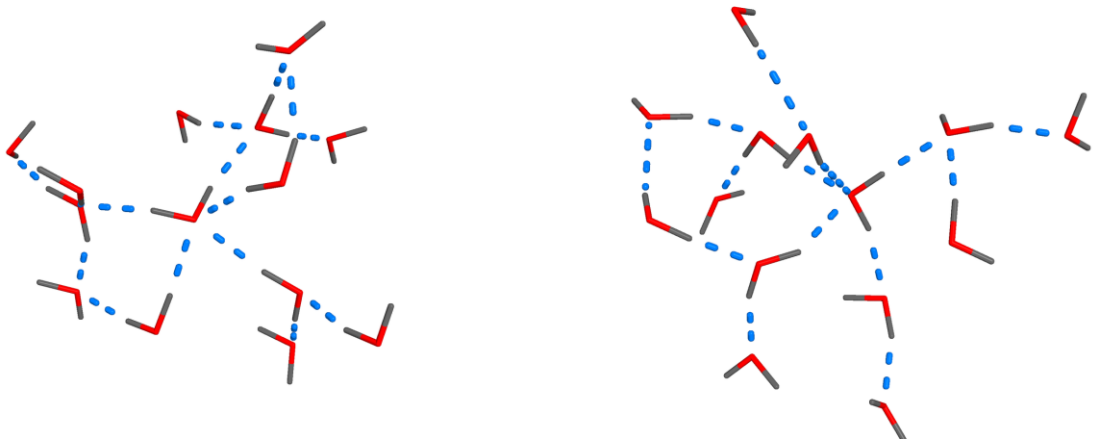


Figure 15: Local hydrogen-bond networks in the objectively selected frames (left: 300 K, frame 7, t = 113.020 ps; right: 273 K, frame 52, t = 127.270 ps). Sticks: oxygen red, hydrogen grey; marine dashes: accepted hydrogen bonds (donor-H···acceptor-O). Illustrative local snapshots.

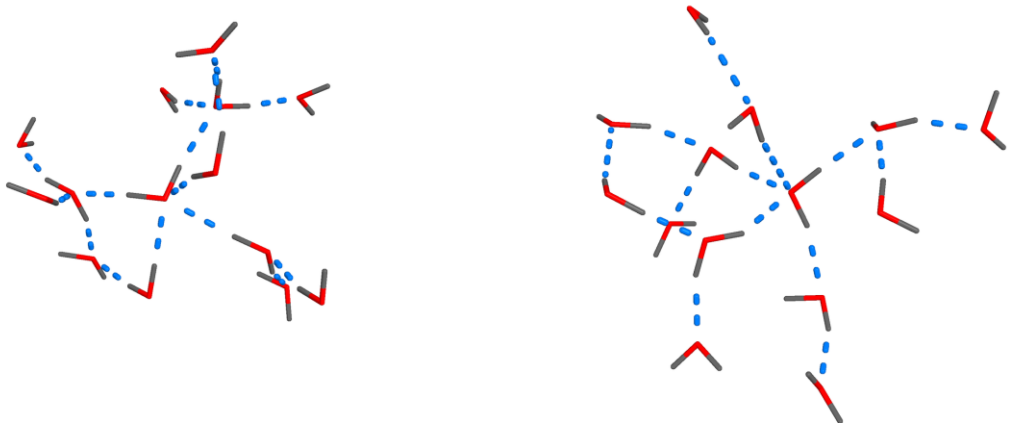


Figure 16: Close-up of the central water and its first hydrogen-bonded shell for the same two frames.

10. Analysis Validation and Repairs

The analysis chain was itself reviewed, and several defects were found in early versions and repaired before any result was accepted. Each repair was made in a new, self-documenting branch; superseded outputs were retained for audit but never reused. The repairs that matter scientifically:

- RDF normalization was corrected so that both branches are normalized against the same ideal-gas reference with exact per-frame volumes, making the 300 K/273 K overlay quantitatively comparable.
- Intramolecular H–H pairs were removed from the H–H RDF (an early version contaminated the intermolecular H–H curve with bonded H–H distances).
- O–H first-feature detection was repaired after review showed the initial peak-locating rule missed the hydrogen-bond contact feature.
- The hydrogen-bond distance histogram range was repaired to cover all accepted 2.4–3.5 Å events without a density-renormalization underflow.

- The tetrahedral-q output schema was repaired so that raw, never-clipped per-water values are stored and pooled.
- Coordination numbers are compared at a matched 3.5 Å radius (and cross-evaluated radii), removing the ambiguity of differently sized first shells.
- The MSD analysis applies a measured centre-of-mass drift correction, unwraps trajectories under periodic boundaries, treats the segmented 273 K production in a restart-safe way (no time origin straddles the segment boundary), and refuses to force a diffusion coefficient: candidate fitting windows were scored against prospective criteria and none was accepted.

All final analyses were independently validated — source-file hashes verified before and after analysis, per-frame hydrogen-bond counts reproduced by an independent code path, distribution and MSD results recomputed and checked, and every packaged figure regenerated from the accepted data. The visualization frames were likewise validated against the accepted time series (the recomputed counts for the two selected frames matched exactly). The result is a two-layer defence: the simulations were validated as they ran, and the analysis of the simulations was validated again afterwards.

11. Discussion

Taken together, the matched comparison draws a coherent picture. Cooling the same TIP3P liquid from a 300 K target to a 273 K target leaves the mean density and box volume very similar, but reorganizes the hydrogen-bond network: roughly 7.9% more geometric hydrogen bonds per frame, bonds that are on average slightly shorter (2.8523 versus 2.8614 Å) and slightly straighter (163.39° versus 162.99°), a higher and sharper O–O first peak at an unchanged 2.775 Å, a more pronounced short-range O–H structural feature in the O–H RDF, a small inward shift of the first H–H feature, a modest rise in tetrahedral order (0.5617 → 0.5908), a small decrease in neighbours within the matched 3.5 Å cutoff (5.2066 → 5.1170; subtle first-shell reorganization) neighbours), and a clearly lower potential energy. Dynamically, the cooler liquid is less mobile at short times, with the oxygen MSD roughly 30% lower at both 1 ps and 2 ps.

The internally consistent interpretation is that the 273 K branch is a more structured liquid: a more populated, more tetrahedrally ordered hydrogen-bond network that still diffuses and fluctuates like a liquid. The lower potential energy is consistent with — but does not by itself measure — stronger network cohesion.

Table 5: Integrated summary of the matched 300 K / 273 K comparison (accepted values; quantities already tabulated with uncertainties in Tables 2–4 are abbreviated here for readability).

Quantity	300 K branch	273 K branch	Direction at 273 K
Sampled temperature	301.79 ± 5.30 K	269.96 ± 4.76 K	cooler, on target
Density	0.976319 g/cm^3	0.976066 g/cm^3	unchanged within noise
Potential energy	−8214.39 kcal/mol	−8518.88 kcal/mol	lower by ≈ 305 kcal/mol
H-bond events/frame	1108.70 ± 17.00	1195.86 ± 17.19	≈ 7.9% more
H-bonds per water	2.5724 ± 0.0394	2.7746 ± 0.0399	more populated
H-bond O···O distance	2.8614 Å	2.8523 Å	slightly shorter
H-bond angle	162.99°	163.39°	slightly straighter

O–O first peak	2.775 Å, g = 2.708	2.775 Å, g = 2.921	sharper, unshifted
Tetrahedral q	0.5617 ± 0.1947	0.5908 ± 0.1922	more tetrahedral
Coordination at 3.5 Å	5.2066	5.1170	small decrease
Oxygen MSD at 1 ps / 2 ps	3.414 / 6.436 $Å^2$	2.380 / 4.345 $Å^2$	less mobile

The boundaries of this interpretation are as important as the findings. No ice formation or phase transition is claimed or observed; the sampled 273 K branch is liquid throughout. No hydrogen-bond energy is measured; geometry is not energy. No rigorous-equilibrium claim is made for these short, single-lineage trajectories, and the noisy NPT pressure is explicitly not used as equilibration evidence. No converged diffusion coefficient is reported, because no fitting window passed the prospective reliability criteria; long-time transport convergence is not demonstrated. The two branches, finally, are related lineages rather than independent replicates, so all conclusions are descriptive of these simulations rather than inferential statements about TIP3P water in general.

12. Limitations

- Short trajectories: each Experiment-2 branch is 5 ps of production; network rearrangements slower than this are invisible.
- Correlated samples: frames are 0.05 ps apart and MSD time origins overlap, so neighbouring frames and lag points are strongly correlated; quoted standard deviations are descriptive spreads, not independent-sample uncertainties.
- No independent replicates: the 273 K branch descends from the 300 K lineage by staged cooling; there are no independent seeds, and no p-values or inferential statistics are reported.
- Finite size: a ~30 Å box (~862 waters) cannot support long-wavelength correlations, and pressure fluctuations are correspondingly large.
- Middle Langevin is the thermostat used for temperature control, and Bussi stochastic cell rescaling is the barostat used for pressure/volume control — the two are distinct components. The Middle Langevin thermostat can affect dynamical transport properties; short-time mobility numbers are therefore protocol-dependent and no thermostat-free transport coefficient is claimed.
- The 273 K production is segmented across a restart; continuity of positions, velocities, and box was validated, but continuity of the thermostat's random-number stream is not claimed.

13. Conclusion

This project delivered a complete, validated molecular dynamics workflow for liquid TIP3P water on consumer hardware: a carefully conditioned, validated, operationally equilibrated 300 K liquid whose thermodynamics, rigid geometry, and structure were verified; and a matched 300 K/273 K comparison in which model, system, protocol, and analysis code were held fixed so that temperature is the intended difference. Within the descriptive limits of these short, correlated trajectories, the final conclusion is that the 273 K branch remained liquid but displayed a more populated and more tetrahedrally ordered hydrogen-bond network — slightly shorter and straighter bonds, sharper radial structure, higher tetrahedral order, lower potential energy, near-identical density — together with reduced short-time oxygen mobility relative to 300 K.

14. Future Work

- Longer production trajectories (hundreds of picoseconds to nanoseconds) to reach the linear MSD regime and enable a defensible, converged self-diffusion coefficient.
- Independent replicate simulations per temperature with different seeds, enabling genuine uncertainty estimates and hypothesis testing.
- Finite-size hydrodynamic corrections and larger simulation boxes for transport properties.
- Comparison across water models (e.g. TIP4P variants) to test model sensitivity of the structural conclusions.
- A dedicated, explicitly framed supercooling / ice-nucleation study as a separate project — none of the present results bear on crystallization.
- In the longer term, machine-learned potentials for water, once the classical workflow is routine.

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